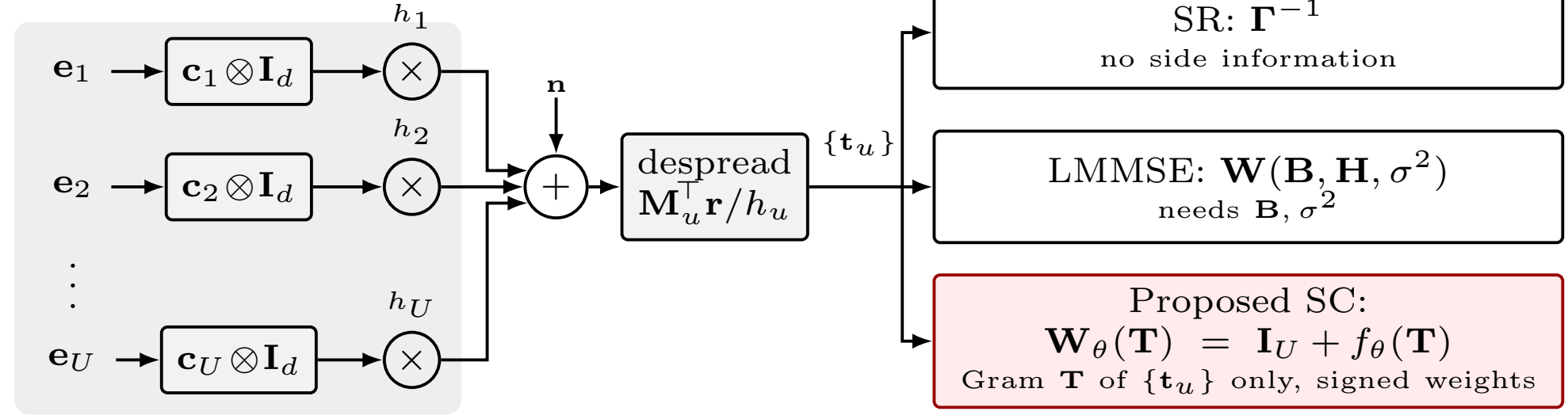

$\mathbf{e}_1$
$\mathbf{c}_1 \otimes \mathbf{I}_d$
h_1
$\mathbf{e}_2$
$\mathbf{c}_2 \otimes \mathbf{I}_d$
h_2
$\mathbf{e}_U$
$\mathbf{c}_U \otimes \mathbf{I}_d$
h_U
$\mathbf{n}$
despread $\mathbf{M}_u^\top \mathbf{r}/h_u$
$\{\mathbf{t}_u\}$
SR: $\mathbf{\Gamma}^{-1}$
no side information
LMMSE: $\mathbf{W}(\mathbf{B}, \mathbf{H}, \sigma^2)$
needs $\mathbf{B}$, σ^2
Proposed SC:
$\mathbf{W}_\theta(\mathbf{T}) = \mathbf{I}_U + f_\theta(\mathbf{T})$
Gram $\mathbf{T}$ of $\{\mathbf{t}_u\}$ only, signed weights
signature spreading, $\mathbf{C}^\top \mathbf{C} = \mathbf{B}$